



TRƯỜNG ĐẠI HỌC
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Đạo đức - Ý chí - Sáng tạo

KHOA CÔNG NGHỆ THÔNG TIN

BỘ MÔN KHOA HỌC DỮ LIỆU

TRÍ TUỆ NHÂN TẠO

LAB 01 - ARTIFICIAL INTELLIGENCE



TS. TRẦN NGỌC VIỆT



HỌC KỲ 3 – NĂM HỌC 2023-2024



KHÓA K28 & K29

THỰC HÀNH 01:

#Game playing

Hai người thay phiên nhau đưa ra các nước đi tuân theo các luật của trò chơi.

- Ví dụ : cờ vua, cờ tướng, cờ ca rô (tic-tắc-toe)

Biểu diễn trong không gian trạng thái, mỗi trạng thái là một tình thế của cuộc chơi

- Trạng thái xuất phát là sự sắp xếp các quân cờ của hai bên khi bắt đầu cuộc chơi
- Các toán tử biến đổi trạng thái là các nước đi hợp lệ
- Các trạng thái kết thúc là các tình thế mà cuộc chơi dừng, được xác định bởi một số điều kiện dừng.
- Hàm kết cuộc: mang giá trị tương ứng với mỗi trạng thái kết thúc.

THỰC HÀNH 01:

Giải thuật minimax

- Chiến lược Minimax

Hai đối thủ trong trò chơi có tên là **MAX** và **MIN**

- Max: biểu diễn cho mục đích của đối thủ này là làm lớn tối đa lợi thế của mình
- Min: biểu diễn cho mục đích của đối thủ này là làm nhỏ tối đa lợi thế của đối phương.

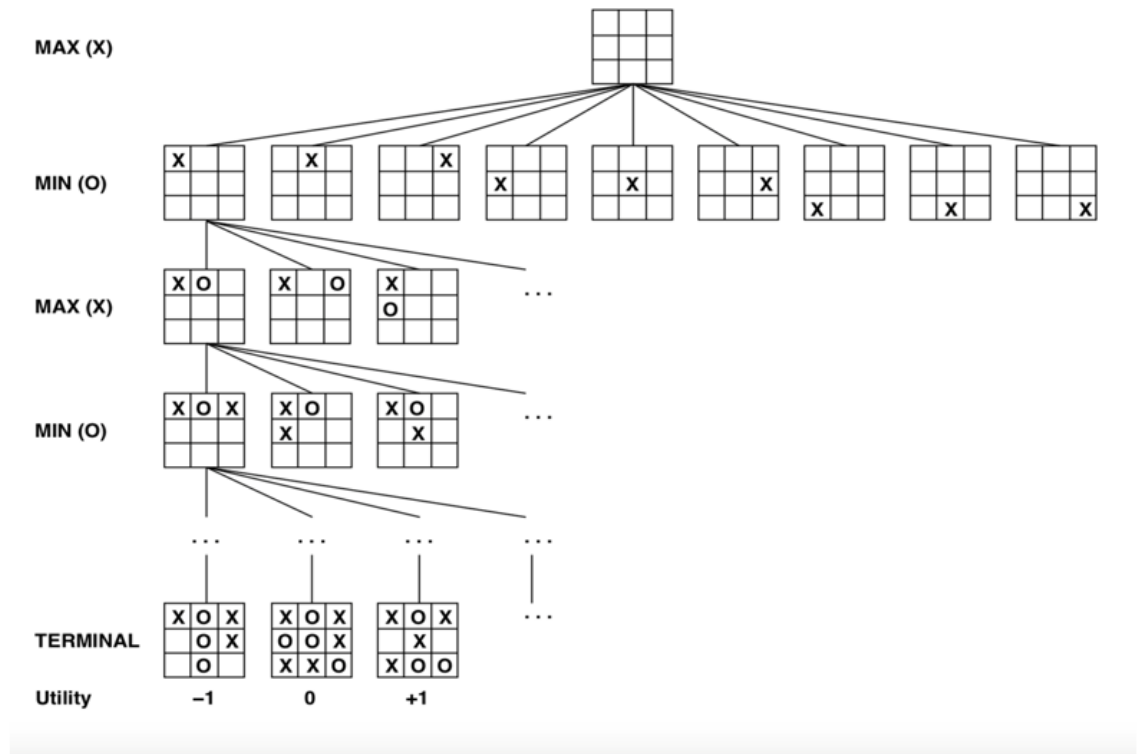
Trên cây tìm kiếm sẽ phân lớp thành các lớp Max và Min.

Với một node **n** bất kỳ,

- Nếu nó thuộc lớp Max thì gán cho nó giá trị Max của các node con
- Nếu nó thuộc lớp Min thì gán cho nó giá trị nhỏ nhất của các node con.

THỰC HÀNH 01:

Giải thuật minimax



ARTIFICIAL INTELLIGENCE

#Thực hành 01: Giải thuật minimax - Game xo 4x4

```
from tkinter import *
from tkinter import messagebox

root = Tk()
root.title('Game XO 4x4')

# Người chơi 1 [X] đi trước, Người chơi 2 [O] tiếp theo sau
clicked = True
count = 0

def disableButtons():
    button1.config(state=DISABLED)
    button2.config(state=DISABLED)
    button3.config(state=DISABLED)
    button4.config(state=DISABLED)
    button5.config(state=DISABLED)
    button6.config(state=DISABLED)
    button7.config(state=DISABLED)
    button8.config(state=DISABLED)
    button9.config(state=DISABLED)
    button10.config(state=DISABLED)
    button11.config(state=DISABLED)
    button12.config(state=DISABLED)
```

```

button13.config(state=DISABLED)

button14.config(state=DISABLED)

button15.config(state=DISABLED)

button16.config(state=DISABLED)

def checkWinner():
    global win
    win = False

    # Cac dieu kien ma nguoi choi 1 [X] chien thang
    if button1["text"] == "X" and button2["text"] == "X" and button3["text"] == "X" and
button4["text"] == "X":

        button1.config(bg="#80ffaa")                #[X][X][X][X]
        button2.config(bg="#80ffaa")                #[O][O][ ][ ]
        button3.config(bg="#80ffaa")                #[ ][ ][O][ ]
        button4.config(bg="#80ffaa")                #[ ][ ][ ][ ]

        win = True
        messagebox.showinfo("OX Game", "Player 1 WINNER!!")

        disableButtons

        start()

    elif button5["text"] == "X" and button6["text"] == "X" and button7["text"] == "X" and
button8["text"] == "X":

```

button5.config(bg="#80ffaa")	#[O][O][O][]
button6.config(bg="#80ffaa")	#[X][X][X][X]
button7.config(bg="#80ffaa")	#[][][][]
button8.config(bg="#80ffaa")	#[][][][]
win = True	
messagebox.showinfo("OX Game", "Player 1 WINNER!!")	
disableButtons	
start()	

elif button9["text"] == "X" and button10["text"] == "X" and button11["text"] == "X" and button12["text"] == "X":

button9.config(bg="#80ffaa")	#[][][][]
button10.config(bg="#80ffaa")	#[O][O][][]
button11.config(bg="#80ffaa")	#[X][X][X][X]
button12.config(bg="#80ffaa")	#[][][][]
win = True	
messagebox.showinfo("OX Game", "Player 1 WINNER!!")	
disableButtons	
start()	

elif button13["text"] == "X" and button14["text"] == "X" and button15["text"] == "X" and button16["text"] == "X":

```

        button13.config(bg="#80ffaa")
        button14.config(bg="#80ffaa")
        button15.config(bg="#80ffaa")
        button16.config(bg="#80ffaa")

        win = True
        messagebox.showinfo("OX Game", "Player 1 WINNER!!")
        disableButtons
        start()

    elif button1["text"] == "X" and button5["text"] == "X" and button9["text"] == "X" and
button13["text"] == "X":
        button1.config(bg="#80ffaa")
        button5.config(bg="#80ffaa")
        button9.config(bg="#80ffaa")
        button13.config(bg="#80ffaa")

        win = True
        messagebox.showinfo("OX Game", "Player 1 WINNER!!")
        disableButtons
        start()

    elif button2["text"] == "X" and button6["text"] == "X" and button10["text"] == "X" and
button14["text"] == "X":
        button2.config(bg="#80ffaa")

```

```
#[ ][ ][ ][ ]
```

```
#[O][O][ ][ ]
```

```
#[ ][ ][ ][ ][ ]
```

```
#[X][X][X][X]
```

```
#[X][O][ ][ ]
```

```
#[X][O][ ][ ]
```

```
#[X][ ][ ][ ]
```

```
#[X][O][ ][ ]
```



```

        button6.config(bg="#80ffaa")
        button10.config(bg="#80ffaa")
        button14.config(bg="#80ffaa")
        win = True
        messagebox.showinfo("OX Game", "Player 1 WINNER!!")
        disableButtons
        start()

    elif button3["text"] == "X" and button7["text"] == "X" and button11["text"] == "X" and
button15["text"] == "X":
        button3.config(bg="#80ffaa")
        button7.config(bg="#80ffaa")
        button11.config(bg="#80ffaa")
        button15.config(bg="#80ffaa")
        win = True
        messagebox.showinfo("OX Game", "Player 1 WINNER!!")
        disableButtons
        start()

    elif button4["text"] == "X" and button8["text"] == "X" and button12["text"] == "X" and
button16["text"] == "X":
        button4.config(bg="#80ffaa")
        button8.config(bg="#80ffaa")

```

```

# [X] [O] [ ]

```

```

# [X] [ ] [ ]

```

```

# [X] [O] [ ]

```

```

# [X] [O] [ ]

```

```

# [X] [ ] [ ]

```

```

# [X] [O] [ ]

```

```

# [X] [O] [ ]

```

```
button12.config(bg="#80ffaa") # [X] [ ] [ ]
```

```
button16.config(bg="#80ffaa")
```

```
win = True
```

```
messagebox.showinfo("OX Game", "Player 1 WINNER!!")
```

```
disableButtons
```

```
start()
```

```
elif button4["text"] == "X" and button7["text"] == "X" and button10["text"] == "X" and  
button13["text"] == "X":
```

```
button4.config(bg="#80ffaa") # [X] [O] [ ]
```

```
button7.config(bg="#80ffaa") # [X] [O] [ ]
```

```
button10.config(bg="#80ffaa") # [X] [ ] [ ]
```

```
button13.config(bg="#80ffaa")
```

```
win = True
```

```
messagebox.showinfo("OX Game", "Player 1 WINNER!!")
```

```
disableButtons
```

```
start()
```

```
# Cac dieu kien nguoi choi 2 [O] chien thang
```

```
if button1["text"] == "O" and button2["text"] == "O" and button3["text"] == "O" and  
button4["text"] == "O":
```

```
button1.config(bg="#80ffaa") # [O] [O] [O] [O]
```

button2.config(bg="#80ffaa")	#[O][O][][]
button3.config(bg="#80ffaa")	#[][][O][]
button4.config(bg="#80ffaa")	#[][][][]
win = True	
messagebox.showinfo("OX Game", "Player 2 WINNER!!")	
disableButtons	
start()	

```
elif button5["text"] == "O" and button6["text"] == "O" and button7["text"] == "O" and
button8["text"] == "O":
```

button5.config(bg="#80ffaa")	#[O][O][O][]
button6.config(bg="#80ffaa")	#[O][O][O][O]
button7.config(bg="#80ffaa")	#[][][][]
button8.config(bg="#80ffaa")	#[][][][]
win = True	
messagebox.showinfo("OX Game", "Player 2 WINNER!!")	
disableButtons	
start()	

```
elif button9["text"] == "O" and button10["text"] == "O" and button11["text"] == "O" and
button12["text"] == "O":
```

button9.config(bg="#80ffaa")	#[][][][]
------------------------------	---------------

button10.config(bg="#80ffaa")	#[O][O][][]
button11.config(bg="#80ffaa")	#[O][O][O][O]
button12.config(bg="#80ffaa")	#[][][][]
win = True	
messagebox.showinfo("OX Game", "Player 2 WINNER!!")	
disableButtons	
start()	

```
elif button13["text"] == "O" and button14["text"] == "O" and button15["text"] == "O" and
button16["text"] == "O":
```

button13.config(bg="#80ffaa")	#[][][][]
button14.config(bg="#80ffaa")	#[O][O][][]
button15.config(bg="#80ffaa")	
#[][][][]	
button16.config(bg="#80ffaa")	#[O][O][O][O]

```
win = True
messagebox.showinfo("OX Game", "Player 2 WINNER!!")
disableButtons
start()
```

```
elif button1["text"] == "O" and button5["text"] == "O" and button9["text"] == "O" and
button13["text"] == "O":
```

button1.config(bg="#80ffaa")	#[O][O][]
button5.config(bg="#80ffaa")	#[O][O][]
button9.config(bg="#80ffaa")	#[O][][]
button13.config(bg="#80ffaa")	
win = True	
messagebox.showinfo("OX Game", "Player 2 WINNER!!")	
disableButtons	
start()	

elif button2["text"] == "O" and button6["text"] == "O" and button10["text"] == "O" and button14["text"] == "O":

button2.config(bg="#80ffaa")	#[O][O][]
button6.config(bg="#80ffaa")	#[O][O][]
button10.config(bg="#80ffaa")	#[O][][]
button14.config(bg="#80ffaa")	
win = True	
messagebox.showinfo("OX Game", "Player 2 WINNER!!")	
disableButtons	
start()	

elif button3["text"] == "O" and button7["text"] == "O" and button11["text"] == "O" and button15["text"] == "O":

button3.config(bg="#80ffaa")	#[O][O][]
button7.config(bg="#80ffaa")	#[O][O][]
button11.config(bg="#80ffaa")	#[O][][]
button15.config(bg="#80ffaa")	
win = True	
messagebox.showinfo("OX Game", "Player 2 WINNER!!")	
disableButtons	
start()	

elif button4["text"] == "O" and button8["text"] == "O" and button12["text"] == "O" and button16["text"] == "O":

button4.config(bg="#80ffaa")	#[O][O][]
button8.config(bg="#80ffaa")	#[O][O][]
button12.config(bg="#80ffaa")	#[O][][]
button16.config(bg="#80ffaa")	
win = True	
messagebox.showinfo("OX Game", "Player 2 WINNER!!")	
disableButtons	
start()	

elif button4["text"] == "O" and button7["text"] == "O" and button10["text"] == "O" and button13["text"] == "O":

```
button4.config(bg="#80ffaa")           #[O][O][ ]
button7.config(bg="#80ffaa")           #[O][O][ ]
button10.config(bg="#80ffaa")          #[O][ ][ ]
button13.config(bg="#80ffaa")
win = True
messagebox.showinfo("OX Game", "Player 2 WINNER!!")
disableButtons
start()
```

Hàm checkDraw nếu hòa nhau

```
def checkDraw():
```

```
    global count, win
```

```
    if count == 16 and win == False:
```

```
        messagebox.showerror("OX Game", "DRAW!!")
```

```
        start()
```

Để xác định các nút mà Người chơi 1 hoặc Người chơi 2 đã nhấp vào

```
def buttonClicked(button):
```

```
    global clicked, count
```

```
    if button["text"] == " " and clicked == True:
```

```
        button["text"] = "X"
        clicked = False
        count += 1
        checkWinner()
        checkDraw()
    elif button["text"] == " " and clicked == False:
        button["text"] = "O"
        clicked = True
        count += 1
        checkWinner()
        checkDraw()
    else:
        messagebox.showerror("OX Game", "LỖI!! Vui lòng chọn lại")

# Để bắt đầu hoặc chơi lại
def start():
    global button1, button2, button3, button4, button5, button6, button7, button8, button9,
    button10, button11, button12, button13, button14, button15, button16
    global clicked, count
    clicked = True
    count = 0
```



```
# Xây dựng các nút cho trò chơi

button1 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button1))

button2 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button2))

button3 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button3))


button4 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button4))

button5 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button5))

button6 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button6))


button7 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button7))

button8 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button8))

button9 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button9))


button10 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button10))

button11 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button11))
```

```
button12 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,  
bg="SystemButtonFace", command=lambda: buttonClicked(button12))
```

```
button13 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,  
bg="SystemButtonFace", command=lambda: buttonClicked(button13))
```

```
button14 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,  
bg="SystemButtonFace", command=lambda: buttonClicked(button14))
```

```
button15 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,  
bg="SystemButtonFace", command=lambda: buttonClicked(button15))
```

```
button16 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,  
bg="SystemButtonFace", command=lambda: buttonClicked(button16))
```

```
# Sắp xếp các nút trên màn hình cho trò chơi
```

```
button1.grid(row=0, column=0)
```

```
button2.grid(row=0, column=1)
```

```
button3.grid(row=0, column=2)
```

```
button4.grid(row=0, column=3)
```

```
button5.grid(row=1, column=0)
```

```
button6.grid(row=1, column=1)
```

```
button7.grid(row=1, column=2)
```

```
button8.grid(row=1, column=3)
```

```
button9.grid(row=2, column=0)
```

```
button10.grid(row=2, column=1)
```

```
button11.grid(row=2, column=2)
```

```
    button12.grid(row=2, column=3)

    button13.grid(row=3, column=0)

    button14.grid(row=3, column=1)

    button15.grid(row=3, column=2)

    button16.grid(row=3, column=3)

# Tạo menu trò chơi
gameMenu = Menu(root)
root.config(menu = gameMenu)

# Tạo menu tùy chọn trò chơi
optionMenu = Menu(gameMenu, tearoff=False)
gameMenu.add_cascade(label="Options", menu=optionMenu)
optionMenu.add_command(label="Restart Game", command=start)

#root.iconbitmap('pyc.ico')

start()

root.mainloop()

-----
```

#Thực hành 02: Giải thuật minimax - Game xo 5x5

#Sinh viên mở rộng từ cấu trúc: if ..elif..

#Sinh viên không được dùng chat GPT

```
from tkinter import *
from tkinter import messagebox

root = Tk()
root.title('OX Game')

# Người chơi 1 [X] đi trước, Người chơi 2 [O] tiếp theo sau
clicked = True
count = 0

def disableButtons():
    button1.config(state=DISABLED)
    button2.config(state=DISABLED)
    button3.config(state=DISABLED)
    button4.config(state=DISABLED)
    button5.config(state=DISABLED)
    button6.config(state=DISABLED)
    button7.config(state=DISABLED)
    button8.config(state=DISABLED)
    button9.config(state=DISABLED)
    button10.config(state=DISABLED)
```

```
button11.config(state=DISABLED)
button12.config(state=DISABLED)
```

```
button13.config(state=DISABLED)
button14.config(state=DISABLED)
button15.config(state=DISABLED)
button16.config(state=DISABLED)
button17.config(state=DISABLED)
button18.config(state=DISABLED)
button19.config(state=DISABLED)
button20.config(state=DISABLED)
button21.config(state=DISABLED)
button22.config(state=DISABLED)
button23.config(state=DISABLED)
button24.config(state=DISABLED)
button25.config(state=DISABLED)
```

```
def checkWinner():
    global win
    win = False
    # Cac dieu kien ma nguoi choi 1 [X] chien thang
```

```

    if button1["text"] == "X" and button2["text"] == "X" and button3["text"] == "X" and
button4["text"] == "X" and button5["text"] == "X":

        button1.config(bg="#80ffaa")                #[X][X][X][X]
        button2.config(bg="#80ffaa")                #[O][O][ ][ ]
        button3.config(bg="#80ffaa")                #[ ][ ][O][ ]
        button4.config(bg="#80ffaa")                #[ ][ ][ ][ ]
        button5.config(bg="#80ffaa")

        win = True

        messagebox.showinfo("OX Game", "Player 1 WINNER!!")

        disableButtons

        start()

```

```

    elif button6["text"] == "X" and button7["text"] == "X" and button8["text"] == "X" and
button9["text"] == "X" and button10["text"] == "X":

        button6.config(bg="#80ffaa")                #[O][O][O ][ ]
        button7.config(bg="#80ffaa")                #[X][X][X][X]
        button8.config(bg="#80ffaa")                #[ ][ ][ ][ ]
        button9.config(bg="#80ffaa")                #[ ][ ][ ][ ]
        button10.config(bg="#80ffaa")

        win = True

        messagebox.showinfo("OX Game", "Player 1 WINNER!!")

        disableButtons

```

```
start()
```

```
elif button11["text"] == "X" and button12["text"] == "X" and button13["text"] == "X" and  
button14["text"] == "X" and button15["text"] == "X":
```

```
    button11.config(bg="#80ffaa")
```

```
    #[ ][ ][ ][ ]
```

```
    button12.config(bg="#80ffaa")
```

```
    #[O][O][ ][ ]
```

```
    button13.config(bg="#80ffaa")
```

```
    #[X][X][X][X]
```

```
    button14.config(bg="#80ffaa")
```

```
    #[ ][ ][ ][ ]
```

```
    button15.config(bg="#80ffaa")
```

```
    win = True
```

```
    messagebox.showinfo("OX Game", "Player 1 WINNER!!")
```

```
    disableButtons
```

```
    start()
```

```
elif button16["text"] == "X" and button17["text"] == "X" and button18["text"] == "X" and  
button19["text"] == "X" and button20["text"] == "X":
```

```
    button16.config(bg="#80ffaa")
```

```
    #[ ][ ][ ][ ]
```

```
    button17.config(bg="#80ffaa")
```

```
    #[O][O][ ][ ]
```

```
    button18.config(bg="#80ffaa")
```

```
    #[ ][ ][ ][ ]
```

```
    button19.config(bg="#80ffaa")
```

```
    #[X][X][X][X]
```

```
    button20.config(bg="#80ffaa")
```

```
    win = True
```

```
messagebox.showinfo("OX Game", "Player 1 WINNER!!")
```

```
disableButtons
```

```
start()
```

```
elif button21["text"] == "X" and button22["text"] == "X" and button23["text"] == "X" and  
button24["text"] == "X" and button25["text"] == "X":
```

```
    button21.config(bg="#80ffaa")
```

```
    #[ ][ ][ ][ ]
```

```
    button22.config(bg="#80ffaa")
```

```
    #[O][O][ ][ ]
```

```
    button23.config(bg="#80ffaa")
```

```
#[[ ][ ][ ][ ]
```

```
    button24.config(bg="#80ffaa")
```

```
    #[X][X][X][X]
```

```
    button25.config(bg="#80ffaa")
```

```
    win = True
```

```
    messagebox.showinfo("OX Game", "Player 1 WINNER!!")
```

```
    disableButtons
```

```
    start()
```

```
elif button1["text"] == "X" and button6["text"] == "X" and button11["text"] == "X" and  
button16["text"] == "X" and button21["text"] == "X":
```

```
    button1.config(bg="#80ffaa")
```

```
    #[X][O][ ][ ]
```

```
    button6.config(bg="#80ffaa")
```

```
    #[X][O][ ][ ]
```

```
    button11.config(bg="#80ffaa")
```

```
    #[X][ ][ ][ ]
```

```
    button16.config(bg="#80ffaa")
```



```
button21.config(bg="#80ffaa")

win = True

messagebox.showinfo("OX Game", "Player 1 WINNER!!")

disableButtons

start()
```

```
elif button2["text"] == "X" and button7["text"] == "X" and button12["text"] == "X" and
button17["text"] == "X" and button22["text"] == "X":
```

```
    button2.config(bg="#80ffaa")
    button7.config(bg="#80ffaa")
    button12.config(bg="#80ffaa")
    button17.config(bg="#80ffaa")
    button22.config(bg="#80ffaa")
    win = True
    messagebox.showinfo("OX Game", "Player 1 WINNER!!")
    disableButtons
    start()
```

```
elif button3["text"] == "X" and button8["text"] == "X" and button13["text"] == "X" and
button18["text"] == "X" and button23["text"] == "X":
```

```
    button3.config(bg="#80ffaa")
    button8.config(bg="#80ffaa")
```

```
button13.config(bg="#80ffaa")
button18.config(bg="#80ffaa")
button23.config(bg="#80ffaa")
win = True
messagebox.showinfo("OX Game", "Player 1 WINNER!!")
disableButtons
start()
```

```
elif button4["text"] == "X" and button9["text"] == "X" and button14["text"] == "X" and
button19["text"] == "X" and button24["text"] == "X":
```

```
button4.config(bg="#80ffaa")
button9.config(bg="#80ffaa")
button14.config(bg="#80ffaa")
button19.config(bg="#80ffaa")
button24.config(bg="#80ffaa")
win = True
messagebox.showinfo("OX Game", "Player 1 WINNER!!")
disableButtons
start()
```

```
elif button5["text"] == "X" and button10["text"] == "X" and button15["text"] == "X" and
button20["text"] == "X" and button25["text"] == "X":
```

```
button5.config(bg="#80ffaa")
button10.config(bg="#80ffaa")
button15.config(bg="#80ffaa")
button20.config(bg="#80ffaa")
button25.config(bg="#80ffaa")
win = True
messagebox.showinfo("OX Game", "Player 1 WINNER!!")
disableButtons
start()
```

```
elif button1["text"] == "X" and button7["text"] == "X" and button13["text"] == "X" and
button19["text"] == "X" and button25["text"] == "X":
```

```
button1.config(bg="#80ffaa")
button7.config(bg="#80ffaa")
button13.config(bg="#80ffaa")
button19.config(bg="#80ffaa")
button25.config(bg="#80ffaa")
win = True
messagebox.showinfo("OX Game", "Player 1 WINNER!!")
disableButtons
start()
```

```
elif button5["text"] == "X" and button9["text"] == "X" and button13["text"] == "X" and  
button17["text"] == "X" and button21["text"] == "X":
```

```
    button5.config(bg="#80ffaa")
```

```
    button9.config(bg="#80ffaa")
```

```
    button13.config(bg="#80ffaa")
```

```
    button17.config(bg="#80ffaa")
```

```
    button21.config(bg="#80ffaa")
```

```
    win = True
```

```
    messagebox.showinfo("OX Game", "Player 1 WINNER!!")
```

```
    disableButtons
```

```
    start()
```

```
# Cac dieu kien nguoi choi 2 [0] chien thang
```

```
if button1["text"] == "O" and button2["text"] == "O" and button3["text"] == "O" and  
button4["text"] == "O" and button5["text"] == "O":
```

```
    button1.config(bg="#80ffaa")
```

```
    button2.config(bg="#80ffaa")
```

```
    button3.config(bg="#80ffaa")
```

```
    button4.config(bg="#80ffaa")
```

```
    button5.config(bg="#80ffaa")
```

```
    win = True
```

```
    messagebox.showinfo("OX Game", "Player 2 WINNER!!")
```

```
    disableButtons
```

```
start()
```

```
elif button6["text"] == "O" and button7["text"] == "O" and button8["text"] == "O" and  
button9["text"] == "O" and button10["text"] == "O":
```

```
    button6.config(bg="#80ffaa")
```

```
    button7.config(bg="#80ffaa")
```

```
    button8.config(bg="#80ffaa")
```

```
    button9.config(bg="#80ffaa")
```

```
    button10.config(bg="#80ffaa")
```

```
    win = True
```

```
    messagebox.showinfo("OX Game", "Player 2 WINNER!!")
```

```
    disableButtons
```

```
    start()
```

```
elif button11["text"] == "O" and button12["text"] == "O" and button13["text"] == "O" and  
button14["text"] == "O" and button15["text"] == "O":
```

```
    button11.config(bg="#80ffaa")
```

```
    button12.config(bg="#80ffaa")
```

```
    button13.config(bg="#80ffaa")
```

```
    button14.config(bg="#80ffaa")
```

```
    button15.config(bg="#80ffaa")
```

```
    win = True
```

```
messagebox.showinfo("OX Game", "Player 2 WINNER!!")
```

```
disableButtons
```

```
start()
```

```
elif button16["text"] == "O" and button17["text"] == "O" and button18["text"] == "O" and  
button19["text"] == "O" and button20["text"] == "O":
```

```
    button16.config(bg="#80ffaa")
```

```
    button17.config(bg="#80ffaa")
```

```
    button18.config(bg="#80ffaa")
```

```
    button19.config(bg="#80ffaa")
```

```
    button20.config(bg="#80ffaa")
```

```
    win = True
```

```
    messagebox.showinfo("OX Game", "Player 2 WINNER!!")
```

```
    disableButtons
```

```
    start()
```

```
elif button21["text"] == "O" and button22["text"] == "O" and button23["text"] == "O" and  
button24["text"] == "O" and button25["text"] == "O":
```

```
    button21.config(bg="#80ffaa")
```

```
    button22.config(bg="#80ffaa")
```

```
    button23.config(bg="#80ffaa")
```

```
    button24.config(bg="#80ffaa")
```

```
button25.config(bg="#80ffaa")

win = True

messagebox.showinfo("OX Game", "Player 2 WINNER!!")

disableButtons

start()
```

```
elif button1["text"] == "O" and button6["text"] == "O" and button11["text"] == "O" and
button16["text"] == "O" and button21["text"] == "O":
```

```
    button1.config(bg="#80ffaa")
    button6.config(bg="#80ffaa")
    button11.config(bg="#80ffaa")
    button16.config(bg="#80ffaa")
    button21.config(bg="#80ffaa")
    win = True
    messagebox.showinfo("OX Game", "Player 2 WINNER!!")
    disableButtons
    start()
```

```
elif button2["text"] == "O" and button7["text"] == "O" and button12["text"] == "O" and
button17["text"] == "O" and button22["text"] == "O":
```

```
    button2.config(bg="#80ffaa")
    button7.config(bg="#80ffaa")
```

```
button12.config(bg="#80ffaa")
button17.config(bg="#80ffaa")
button22.config(bg="#80ffaa")
win = True
messagebox.showinfo("OX Game", "Player 2 WINNER!!")
disableButtons
start()
```

```
elif button3["text"] == "O" and button8["text"] == "O" and button13["text"] == "O" and
button18["text"] == "O" and button23["text"] == "O":
```

```
    button3.config(bg="#80ffaa")
    button8.config(bg="#80ffaa")
    button13.config(bg="#80ffaa")
    button18.config(bg="#80ffaa")
    button23.config(bg="#80ffaa")
    win = True
    messagebox.showinfo("OX Game", "Player 2 WINNER!!")
    disableButtons
    start()
```

```
elif button4["text"] == "O" and button9["text"] == "O" and button14["text"] == "O" and
button19["text"] == "O" and button24["text"] == "O":
```



```
button4.config(bg="#80ffaa")
button9.config(bg="#80ffaa")
button14.config(bg="#80ffaa")
button19.config(bg="#80ffaa")
button24.config(bg="#80ffaa")

win = True

messagebox.showinfo("OX Game", "Player 2 WINNER!!")

disableButtons

start()
```

```
elif button5["text"] == "O" and button10["text"] == "O" and button15["text"] == "O" and
button20["text"] == "O" and button25["text"] == "O":
```

```
button5.config(bg="#80ffaa")
button10.config(bg="#80ffaa")
button15.config(bg="#80ffaa")
button20.config(bg="#80ffaa")
button25.config(bg="#80ffaa")

win = True

messagebox.showinfo("OX Game", "Player 2 WINNER!!")

disableButtons

start()
```

```
elif button1["text"] == "O" and button7["text"] == "O" and button13["text"] == "O" and  
button19["text"] == "O" and button25["text"] == "O":
```

```
    button1.config(bg="#80ffaa")
```

```
    button7.config(bg="#80ffaa")
```

```
    button13.config(bg="#80ffaa")
```

```
    button19.config(bg="#80ffaa")
```

```
    button25.config(bg="#80ffaa")
```

```
    win = True
```

```
    messagebox.showinfo("OX Game", "Player 2 WINNER!!")
```

```
    disableButtons
```

```
    start()
```

```
elif button5["text"] == "O" and button9["text"] == "O" and button13["text"] == "O" and  
button17["text"] == "O" and button21["text"] == "O":
```

```
    button5.config(bg="#80ffaa")
```

```
    button9.config(bg="#80ffaa")
```

```
    button13.config(bg="#80ffaa")
```

```
    button17.config(bg="#80ffaa")
```

```
    button21.config(bg="#80ffaa")
```

```
    win = True
```

```
    messagebox.showinfo("OX Game", "Player 2 WINNER!!")
```

```
    disableButtons
```

```

start()

# Hàm checkDraw nếu hòa nhau
def checkDraw():
    global count, win

    if count == 25 and win == False:
        messagebox.showerror("OX Game", "DRAW!!")
        start()

# Để xác định các nút mà Người chơi 1 hoặc Người chơi 2 đã nhấp vào
def buttonClicked(button):
    global clicked, count

    if button["text"] == " " and clicked == True:
        button["text"] = "X"
        clicked = False
        count += 1
        checkWinner()
        checkDraw()

    elif button["text"] == " " and clicked == False:
        button["text"] = "O"
        clicked = True
        count += 1

```

```

        checkWinner()

        checkDraw()

    else:

        messagebox.showerror("OX Game", "LỖI!! Vui lòng chọn lại")

# Để bắt đầu hoặc chơi lại

def start():

    global button1, button2, button3, button4, button5, button6, button7, button8, button9,
    button10, button11, button12, button13, button14, button15, button16, button17, button18, button19,
    button20, button21, button22, button23, button24, button25

    global clicked, count

    clicked = True

    count = 0

    # Xây dựng các nút cho trò chơi

    button1 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button1))

    button2 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button2))

    button3 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button3))

    button4 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button4))

    button5 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button5))

    button6 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button6))

```

```
button7 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button7))

button8 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button8))

button9 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button9))

button10 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button10))

button11 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button11))

button12 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button12))

button13 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button13))

button14 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button14))

button15 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button15))

button16 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button16))

button17 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button17))

button18 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button18))

button19 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button19))
```

```
button20 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button20))

button21 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button21))

button22 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button22))

button23 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button23))

button24 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button24))

button25 = Button(root, text=" ", font=("Helvetica, 20"), height=3, width=7,
bg="SystemButtonFace", command=lambda: buttonClicked(button25))
```

```
# Sắp xếp các nút trên màn hình cho trò chơi
```

```
button1.grid(row=0, column=0)
button2.grid(row=0, column=1)
button3.grid(row=0, column=2)
button4.grid(row=0, column=3)
button5.grid(row=0, column=4)
button6.grid(row=1, column=0)
button7.grid(row=1, column=1)
button8.grid(row=1, column=2)
button9.grid(row=1, column=3)
button10.grid(row=1, column=4)
```

```
button11.grid(row=2, column=0)
button12.grid(row=2, column=1)
button13.grid(row=2, column=2)
button14.grid(row=2, column=3)
button15.grid(row=2, column=4)
button16.grid(row=3, column=0)
button17.grid(row=3, column=1)
button18.grid(row=3, column=2)
button19.grid(row=3, column=3)
button20.grid(row=3, column=4)
button21.grid(row=4, column=0)
button22.grid(row=4, column=1)
button23.grid(row=4, column=2)
button24.grid(row=4, column=3)
button25.grid(row=4, column=4)

# Tạo menu trò chơi
gameMenu = Menu(root)
root.config(menu = gameMenu)

# Tạo menu tùy chọn trò chơi
optionMenu = Menu(gameMenu, tearoff=False)
gameMenu.add_cascade(label="Options", menu=optionMenu)
```

```
optionMenu.add_command(label="Restart Game", command=start)
#root.iconbitmap('pyc.ico')
start()
root.mainloop()
```

#Thực hành 03: Giải thuật minimax - Game xo 6x6 (mở rộng)

#Sinh viên mở rộng từ cấu trúc: if ..elif..

#Sinh viên không được dùng chat GPT

. . .

#Thực hành 04: Giải thuật minimax - Game xo 8x8

#Sinh viên mở rộng từ cấu trúc: if ..elif..

#Sinh viên không được dùng chat GPT

. . .

